

So, the Domain Controller is healthy. I currently have: Active Directory installed, DNS working, Organizational Units, Users, Global Groups, Domain Local Groups, AGDLP implemented, Department folders, and initial NTFS permissions.

The next question I want to ask is: How do employees actually use this environment?

I need to build the Windows 11 Client VM. It is now downloaded, but quite slow cause of the USB....sad.

Once getting into the Client VM, we want to rename the computer.

Rename-Computer -NewName "CLIENT01" -Restart

After restart we can then verify this.

Hostname and **whoami**.

Results: CLIENT01 and CLIENT01/adminpc

Understanding the Client: running **ipconfig /all**

Looking at IPv4 Address, DNS Server, Default Gateway, DHCP Enabled, Adapter

Trying to ping the DC with its IP address from the Client VM. **Connectivity worked!**

Now pinging using (SRV-DC01). **Worked!**

Try: **nslookup kevinlab.local**

Remember: nslookup asks a DNS server: What IP address belongs to kevinlab.local?

The command is currently failing. I need to make the Windows 11 client use the Windows Server DC for DNS, instead of my home router.

The IP address is: 192.168.1.196

On the Windows 11 VM: Windows + R and type ncpa.cpl. Right click Ethernet and Select Properties. Double click Internet Protocol Version 4 and Keep the IP section as it is. Use the DNS server addresses:

Preferred DNS Server: Windows Server IP

Alternate DNS Server: Blank

Using commands ipconfig /flushdns and ipconfig /registerdns to clear the old DNS information.

The nslookup kevinlab.local command now works.

Since we have now configured Windows, I am now creating a snapshot. CLIENT01-Baseline.

Some review questions:

What is a Domain Controller?

A Domain Controller is a server that manages an organization's Windows domain. So, in my lab the Windows Server VM is the Domain Controller for: kevinlab.local. It stores and manages

things such as: user accounts, password information, computers joined to the domain, security groups, group policy, and permissions and access rules.

What is a Client?

A client is a computer that uses services provided by a server. In my lab, the Windows 11 VM is the client.

Authentication vs Authorization

Authentication: Who are you?

Authorization: What are you allowed to do?

Why does a client use the Domain Controller's DNS?

Active Directory depends heavily on DNS. The client does not only need to resolve kevinlab.local, but it also needs to locate specific Active Directory services.

Why do companies centralize authentication?

Without centralized authentication, administrators might need to create and remove separate local accounts on hundreds of individual computers. That becomes difficult to secure and maintain.

Going to continue now with the Domain Join and Authentication.

Just want to continue the session by confirming the DNS server is working, using ipconfig /all.

I also want to confirm that VMware Tools is downloaded. I will check using the command:

Get-Service VMTools -ErrorAction SilentlyContinue

Nothing appeared so I am going to attempt to fix this.

Next I will Verify Active Directory Discovery. I want to prove the client can locate the actual AD services.

Run the command: **nslookup -type=SRV _ldap._tcp.dc._msdcs.kevinlab.local**

Nslookup queries a DNS server. -type=SRV tells nslookup to search for an SRV record instead of a normal host/IP record. (SRV record identifies what service is available, which server provides it, which port it uses, and its priority and weight), _ldap is the service being requested, which is LDAP. LDAP is the protocol clients use to communicate with AD and search directory information. _tcp is the LDAP service available over TCP. dc means we are looking specifically for a Domain Controller. _msdcs means: Microsoft Domain Controller Services. It is a special DNS area used by AD to help computers locate Domain Controllers and other AD services.

Basically: this asks DNS: Which DC provides LDAP services for this domain? In the response I see the hostname to answer this question is "srv-dc01.kevinlab.local"

Now we will run the command: **nltest /dsgetdc:kevinlab.local**.

This command asks Windows to find a domain controller for the kevinlab.local domain and show information about it. Nltest is used to test and troubleshoot: Domain Controllers, Domain trust relationships, secure channels, and domain discovery. /dsgetdc: "Get a Domain Controller for this domain."

Results of command:

DC name: SRV-DC01.kevinlab.local

DC address: displayed there

Domain name: kevinlab.local

Forest name: ^^

Flags displayed:

Now, we are finally joining CLIENT01 to the domain.

Add-Computer `

- Domain Name "kevinlab.local" `

- Credential "KEVINLAB\Administrator" `

-Restart

A credential window appears after using this command.

Before this: CLIENT01 was an independent Windows computer. Now, CLIENT01 becomes a trusted computer object inside kevinlab.local.

We now need to prove that the Domain Join Worked. Going to log in with adminpc, which is the local administrator, but there is also an option to sign in with Other User now.

To check domain membership: **Get-CimInstance Win32_ComputerSystem | Select-Object Name, Domain, PartOfDomain**

Name: CLIENT01

Domain: kevinlab.local

PartOfDomain: True

Now, lets move CLIENT01 into the correct OU. This will be done on the Server now.

Server Manager -> Tools -> Active Directory Users and Computers

By default, CLIENT01 is inside Computers. I am moving it into

KEVINLAB-Computers/Workstations.

We must move it as the default Computers container cannot be used effectively for targeted Group Policy management. Placing workstations into a proper OU allows policies to be applied specifically to employee endpoints.

To verify with powershell: **Get-ADComputer CLIENT01 | Select-Object Name, DistinguishedName**

Now we will perform basic Domain User Authentication. Sign out of CLIENT01. Log in with one of the other users created, preferably ahr.

This put me through the “Hi, Getting Things Ready For You, This might take a few minutes,” menu. First time logging into Alice’s account.

To verify: **whoami** and **hostname**.

Identify the authenticating DC: **\$env:LOGONSERVER** (SRV-DC01 returned)

To inspect Kerberos tickets: **klist**, look for the ticket associated with krbtgt/KEVINLAB.LOCAL

Inspecting applied policy information: **gpresult /r**

I don’t have custom GPOs yet, but command still shows user, computer, domain membership, and applied policy information.